Examiner only

4. Enzymes are involved in all metabolic reactions.

(a) (i) State the **type** of biological molecule that all enzymes are made of. [1]

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Enzymes can break down larger molecules into smaller units in the digestive system.

mouth	amylase	lipase	liver
large intestine	amino acids	fibre	stomach

Enzyme	Larger molecules	Smaller units	Site of action
Carbohydrase	Starch	Glucose	.... and small intestine
....	Fats and oils	Fatty acids and glycerol	small intestine
Protease	Protein	....	.... and small intestine

(ii) **Complete the table** using only the words in the box. [4]

(b) Students investigated the effect of temperature on the digestion of starch.

The students mixed starch and amylase (a carbohydrase) at different temperatures.

They used iodine to test for the presence of starch. Iodine changes colour from brown to blue-black if starch is present.

The results of the students' investigation are shown below.

Temperature of the solution (°C)	Colour of the solution					
	at start	after 1 min	after 2 min	after 3 min	after 4 min	after 5 min
20	blue-black	blue-black	blue-black	blue-black	blue-black	blue-black
30	blue-black	blue-black	blue-black	blue-black	blue-black	brown
40	blue-black	brown	brown	brown	brown	brown
50	blue-black	blue-black	brown	brown	brown	brown
60	blue-black	blue-black	blue-black	brown	brown	brown
70	blue-black	blue-black	blue-black	blue-black	brown	brown
80	blue-black	blue-black	blue-black	blue-black	blue-black	blue-black

(i) Describe how temperature affects the digestion of starch between **20 °C** and **70 °C**. [3]

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(ii) Explain the results at **80 °C**. [2]

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Examiner only

(c) When starch is completely broken down by amylase the final product is glucose.

(i) Describe the test that the students need to perform to show the presence of glucose. [2]

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(ii) State the colour **change** that shows that glucose is present. [1]

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